

Security+ 1.4 · Encrypting Data

Mestre CyberSec · Lesson 1.4 — Encrypting Data (9:48) · [Assistir Vídeo no YouTube](#) · Treino Interativo: edusouzamarques.github.io/cybersec

Executive Lesson Summary: Dado em 3 ESTADOS: AT REST (parado — BitLocker, TDE no banco), IN TRANSIT (viajando — TLS/VPN), IN USE (sendo processado — confidential computing, enclaves CPU). Block cipher (AES — cifra chunks de 128 bits) vs Stream cipher (byte a byte, rápido).

💡 Key Exam Highlights & Critical Concepts

- At rest = disco; In transit = rede (TLS); In use = memória (mais difícil)
- AES block cipher cifra em blocos de 128 bits
- Stream cipher: ChaCha20 — bom pra celular, sem hardware AES

🎯 Exam Tips & Typical CompTIA Scenarios

- Na prova: 'data stored in the database' = at rest; 'during network transfer' = in transit
- Pergunta de 'which provides protection for data in USE' = confidential computing

Beyond the Video: Official Exam Concepts, Protocols & Traps (Section 1.4 — Encrypting Data)

Why is this advanced coverage required for SY0-701? Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

• Data at Rest

Data stored on a physical medium (disk, database, tape, USB) that is not currently moving or being processed; protected by full-disk encryption like BitLocker or FileVault, or database-level encryption like TDE.

 **Exam Trap:** Gatilho: 'data stored in the database', 'files on a laptop', 'backup tape' = AT REST. Soluções típicas: BitLocker, FileVault, TDE (Transparent Data Encryption). Não confunda com tráfego de rede.

• Data in Transit

Data actively moving across a network between systems; protected by transport encryption such as TLS/HTTPS, IPsec VPN, or SSH to prevent interception and man-in-the-middle attacks.

 **Exam Trap:** Gatilho: 'during network transfer', 'data sent over the internet', 'web browser to server' = IN TRANSIT. Protocolos-chave: TLS, HTTPS, IPsec, SSH. Se aparecer 'emails being sent' pense em S/MIME ou TLS de transporte.

• Data in Use

Data being actively processed in memory or CPU; the hardest state to protect because encryption must persist during computation, requiring confidential computing, secure enclaves, or CPU-based trusted execution environments.

 **Exam Trap:** Gatilho de prova: 'which provides protection for data IN USE' = CONFIDENTIAL COMPUTING / secure enclave (Intel SGX, AMD SEV, ARM TrustZone). Memória RAM é volátil mas vulnerável — não confunda com at rest.

• Block Cipher

Encryption algorithm that processes data in fixed-size chunks (commonly 128 bits) at a time using a symmetric key; AES is the modern standard, with DES and 3DES as legacy examples.

 **Exam Trap:** Gatilho: 'encrypts 128-bit blocks' = BLOCK CIPHER. AES = block cipher de 128 bits (chaves 128/192/256). Precisa de padding se os dados não preencherem o bloco. Diferencie de stream cipher, que cifra byte a byte.

• Stream Cipher

Encryption algorithm that encrypts data one byte (or bit) at a time by combining plaintext with a keystream; ChaCha20 and RC4 are examples, offering speed without requiring AES hardware acceleration.

 **Exam Trap:** Gatilho: 'good for mobile devices', 'no AES hardware acceleration', 'encrypts one byte at a time' = STREAM CIPHER (ChaCha20). RC4 é stream cipher fraco e depreciado — não use em TLS moderno.

• Confidential Computing

Security model that protects data in use by performing computation inside a hardware-based trusted execution environment (secure enclave), so even the OS or hypervisor cannot read the data being processed.

 **Exam Trap:** Se a pergunta for 'protection for data IN USE', a resposta é confidential computing / secure enclave (Intel SGX, AMD SEV, ARM TrustZone). Chave: isolamento em nível de hardware de memória em processamento.

• Transparent Data Encryption (TDE)

Database-level encryption that encrypts data at rest (data files, logs, backups) transparently to applications, requiring no changes to the application and unlocking data only when needed by the DB engine.

 **Exam Trap:** Gatilho: 'encrypting database at rest without changing the app' = TDE. É at REST, não in transit. Diferencie de column-level encryption, que é mais granular.

• Transport Layer Security (TLS)

Cryptographic protocol that provides confidentiality and integrity for data in transit over a network, using symmetric encryption (AES or ChaCha20) after an asymmetric handshake; successor to SSL and foundation of HTTPS.

⚠ **Exam Trap:** Gatilho: 'protect data in transit', 'HTTPS', 'secure web traffic' = TLS. Versões modernas: TLS 1.2/1.3. SSL está depreciado. Usa criptografia híbrida: assimétrica no handshake, simétrica nos dados.

✓ Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

QUESTION 1 · EXAM SCENARIO

A hospital runs a SQL Server database that stores patient records. Compliance requires that database files and backups be encrypted at rest without changing application queries or connection strings. Which control should the security administrator implement?

- A) Transparent Data Encryption on the database
- B) TLS 1.3 for all client connections
- C) Confidential computing enclaves on the database server
- D) IPsec between the application and database servers

QUESTION 2 · EXAM SCENARIO

Remote employees use public Wi-Fi to access a cloud CRM. The security team must protect credentials and session data as they travel between the browser and the cloud service. Which control best addresses this requirement?

- A) Full-disk encryption on employee laptops
- B) TLS for the CRM web application
- C) Transparent Data Encryption in the CRM database
- D) Confidential computing enclaves in the cloud

QUESTION 3 · EXAM SCENARIO

A company issues laptops to engineers who store source code and design documents locally. After several laptops are stolen from vehicles, management wants to prevent an attacker who removes the drive from reading the data. Which solution should be implemented?

- A) TLS for all internal web applications
- B) Transparent Data Encryption on file servers
- C) Full-disk encryption on the laptops
- D) Confidential computing for developer workloads

QUESTION 4 · EXAM SCENARIO

A payroll application runs in a public cloud. The legal team is concerned that cloud provider administrators could inspect payroll data while it is being processed in memory. Which technology should be used to protect the data in use?

- A) TLS between the payroll users and the cloud
- B) Transparent Data Encryption on the payroll database
- C) IPsec between the corporate network and the cloud
- D) Confidential computing with hardware-based trusted execution environments

QUESTION 5 · EXAM SCENARIO

A security engineer must encrypt a real-time voice stream where each byte should be protected immediately with minimal buffering and latency. Which cipher type is best suited for this requirement?

- A) Block cipher with PKCS#7 padding
- B) Stream cipher
- C) Asymmetric cipher such as RSA
- D) Cryptographic hash function

QUESTION 6 · EXAM SCENARIO

A developer selects AES-256-GCM to encrypt a multi-terabyte backup file. The design document must correctly classify the underlying cryptographic algorithm. Which statement is accurate?

- A) AES is a stream cipher because GCM uses a counter mode.
- B) AES is a symmetric block cipher that encrypts 128-bit blocks, and GCM provides authenticated encryption.
- C) AES is an asymmetric cipher because it uses Galois field multiplication.
- D) AES is a hash function because GCM produces an authentication tag.

QUESTION 7 · EXAM SCENARIO

A branch office connects to headquarters over the public internet using an IPsec VPN. The security manager asks which data state the VPN primarily protects for files being transferred between the sites. What is the correct answer?

- A) Data at rest
- B) Data in transit
- C) Data in use
- D) Data in all three states simultaneously

QUESTION 8 · EXAM SCENARIO

An attacker steals encrypted backup tapes from an offsite storage vault. The organization already uses TLS for database connections and enclaves for data processing. Which additional control most directly protects the data on the stolen tapes?

- A) Transparent Data Encryption for the database backups
- B) TLS 1.3 for backup server administration
- C) Confidential computing enclaves on the backup server
- D) Multifactor authentication for backup administrators

QUESTION 9 · EXAM SCENARIO

A financial company must encrypt high-volume traffic between microservices. The servers include AES-NI hardware acceleration, and the design requires both confidentiality and integrity. Which cryptographic approach is most appropriate?

- A) AES-GCM
- B) RSA-4096
- C) SHA-256
- D) AES-ECB without authentication

QUESTION 10 · EXAM SCENARIO

A cloud security architect is mapping controls to protect customer data in all three states. Which mapping correctly aligns a control with each state?

- A)** Full-disk encryption for data at rest, TLS for data in transit, confidential computing for data in use
- B)** TLS for data at rest, full-disk encryption for data in transit, TDE for data in use
- C)** TDE for data in use, IPsec for data at rest, CPU enclaves for data in transit
- D)** Confidential computing for data at rest, TLS for data in use, TDE for data in transit

ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

Question 1: Option A — Transparent Data Encryption on the database

Transparent Data Encryption (TDE) encrypts database files and backups at rest while keeping encryption transparent to applications. TLS and IPsec protect data in transit, and confidential computing protects data in use.

Question 2: Option B — TLS for the CRM web application

TLS protects data in transit between the browser and the cloud service by encrypting and authenticating the communication channel. Full-disk encryption and TDE protect data at rest, while enclaves protect data in use.

Question 3: Option C — Full-disk encryption on the laptops

Full-disk encryption protects data at rest on the entire laptop drive, so removing the drive does not expose readable data. TLS protects data in transit, TDE protects database files, and confidential computing protects data in use.

Question 4: Option D — Confidential computing with hardware-based trusted execution environments

Confidential computing uses trusted execution environments, such as CPU enclaves and memory encryption, to protect data while it is being processed. TLS and IPsec protect data in transit, and TDE protects data at rest.

Question 5: Option B — Stream cipher

A stream cipher encrypts data one byte at a time and is well suited for real-time streams that cannot wait for a full block. Block ciphers process fixed-size chunks and may add padding; asymmetric ciphers and hashes do not provide symmetric stream encryption.

Question 6: Option B — AES is a symmetric block cipher that encrypts 128-bit blocks, and GCM provides authenticated encryption.

AES is a symmetric block cipher that processes fixed 128-bit blocks. GCM is a mode of operation that turns AES into an authenticated encryption mode, but the underlying AES algorithm remains a block cipher.

Question 7: Option B — Data in transit

An IPsec VPN encrypts traffic as it travels across the public internet, so it primarily protects data in transit. Data at rest requires controls such as full-disk or database encryption, and data in use requires controls such as confidential computing.

Question 8: Option A — Transparent Data Encryption for the database backups

Transparent Data Encryption can encrypt database backups at rest, so stolen tapes contain ciphertext rather than readable data. TLS, enclaves, and MFA do not protect data at rest on physical media.

Question 9: Option A — AES-GCM

AES-GCM is an authenticated encryption mode that provides confidentiality and integrity, and it benefits from AES-NI hardware acceleration. RSA is asymmetric and slower for bulk traffic, SHA-256 is a hash and does not encrypt, and AES-ECB lacks integrity and is insecure for many uses.

Question 10: Option A — Full-disk encryption for data at rest, TLS for data in transit, confidential computing for data in use

Full-disk encryption and TDE protect data at rest, TLS and IPsec protect data in transit, and confidential computing with CPU enclaves protects data in use. The other mappings assign controls to the wrong data state.